

Systems of Linear Equations: Worked Examples and Word Problems (GED)

GED Mathematics

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1 Introduction

1.1 What Are Systems of Linear Equations?

A system of linear equations is a set of two or more linear equations with two or more variables. The solution to a system is the set of values that satisfies all equations simultaneously. On the GED, you need to find the point(s) where two or more lines intersect.

1.2 Three Main Solution Methods

There are three primary methods for solving systems of linear equations:

- **Substitution Method:** Solve one equation for a variable and substitute it into the other equation
- **Elimination Method:** Multiply one or both equations by constants and add or subtract them to eliminate a variable
- **Graphing Method:** Graph both equations and find the intersection point

1.3 Why These Skills Matter for GED Success

Solving systems of linear equations is essential for:

- Real-world problem solving involving multiple constraints
- Understanding relationships between variables
- Developing algebraic reasoning skills
- Preparing for college-level mathematics

2 10 Worked Examples

2.1 Examples 1–3: Substitution Method

2.1.1 Example 1

System:

$$\begin{aligned}y &= 2x + 1 \\x + y &= 7\end{aligned}$$

Solution:

1. The first equation already gives us y in terms of x : $y = 2x + 1$
2. Substitute this into the second equation:

$$x + (2x + 1) = 7$$

3. Combine like terms:

$$3x + 1 = 7$$

4. Solve for x :

$$3x = 6$$

$$x = 2$$

5. Substitute back to find y :

$$y = 2(2) + 1 = 5$$

Answer: $x = 2, y = 5$

2.1.2 Example 2

System:

$$x = y + 3$$

$$2x + y = 12$$

Solution:

1. The first equation gives us x in terms of y : $x = y + 3$
2. Substitute into the second equation:

$$2(y + 3) + y = 12$$

3. Distribute and combine like terms:

$$2y + 6 + y = 12$$

$$3y + 6 = 12$$

4. Solve for y :

$$3y = 6$$

$$y = 2$$

5. Substitute back to find x :

$$x = 2 + 3 = 5$$

Answer: $x = 5, y = 2$

2.1.3 Example 3

System:

$$\begin{aligned}y &= 3x - 2 \\2x + 3y &= 16\end{aligned}$$

Solution:

1. The first equation gives us $y = 3x - 2$
2. Substitute into the second equation:

$$2x + 3(3x - 2) = 16$$

3. Distribute and combine like terms:

$$\begin{aligned}2x + 9x - 6 &= 16 \\11x - 6 &= 16\end{aligned}$$

4. Solve for x :

$$\begin{aligned}11x &= 22 \\x &= 2\end{aligned}$$

5. Substitute back to find y :

$$y = 3(2) - 2 = 4$$

Answer: $x = 2, y = 4$

2.2 Examples 4–6: Elimination Method

2.2.1 Example 4

System:

$$\begin{aligned}2x + y &= 8 \\x - y &= 1\end{aligned}$$

Solution:

1. Notice that the y coefficients are opposites ($+1$ and -1). Add the equations:

$$\begin{aligned}(2x + y) + (x - y) &= 8 + 1 \\3x &= 9 \\x &= 3\end{aligned}$$

2. Substitute $x = 3$ into the second equation:

$$\begin{aligned}3 - y &= 1 \\-y &= -2 \\y &= 2\end{aligned}$$

Answer: $x = 3, y = 2$

2.2.2 Example 5

System:

$$3x + 2y = 12$$

$$2x - 2y = 8$$

Solution:

1. The y coefficients are opposites (+2 and -2). Add the equations:

$$(3x + 2y) + (2x - 2y) = 12 + 8$$

$$5x = 20$$

$$x = 4$$

2. Substitute $x = 4$ into the first equation:

$$3(4) + 2y = 12$$

$$12 + 2y = 12$$

$$2y = 0$$

$$y = 0$$

Answer: $x = 4, y = 0$

2.2.3 Example 6

System:

$$4x + 3y = 18$$

$$2x + 3y = 12$$

Solution:

1. The y coefficients are the same (+3). Subtract the second equation from the first:

$$(4x + 3y) - (2x + 3y) = 18 - 12$$

$$2x = 6$$

$$x = 3$$

2. Substitute $x = 3$ into the second equation:

$$2(3) + 3y = 12$$

$$6 + 3y = 12$$

$$3y = 6$$

$$y = 2$$

Answer: $x = 3, y = 2$

2.3 Examples 7–8: Graphing Method

2.3.1 Example 7

System:

$$y = x + 1$$

$$y = -x + 5$$

Solution:

1. For graphing, we find where the two lines intersect. Set the equations equal:

$$x + 1 = -x + 5$$

2. Solve for x :

$$2x = 4$$

$$x = 2$$

3. Substitute back into either equation:

$$y = 2 + 1 = 3$$

4. The lines intersect at the point $(2, 3)$

Answer: $x = 2, y = 3$

2.3.2 Example 8

System:

$$y = 2x - 1$$

$$y = -x + 5$$

Solution:

1. Set the equations equal:

$$2x - 1 = -x + 5$$

2. Solve for x :

$$3x = 6$$

$$x = 2$$

3. Substitute back into either equation:

$$y = 2(2) - 1 = 3$$

4. The lines intersect at $(2, 3)$

Answer: $x = 2, y = 3$

2.4 Examples 9–10: Special Cases

2.4.1 Example 9: No Solution (Parallel Lines)

System:

$$y = 2x + 3$$

$$y = 2x + 1$$

Solution:

1. Both equations have the same slope ($m = 2$) but different y -intercepts ($b = 3$ and $b = 1$)
2. These are parallel lines that never intersect
3. Attempting to solve algebraically:

$$2x + 3 = 2x + 1$$

$$3 = 1$$

4. This is a contradiction, confirming there is no solution

Answer: **No solution** (parallel lines do not intersect)

2.4.2 Example 10: Infinitely Many Solutions (Identical Lines)

System:

$$2x + 4y = 8$$

$$x + 2y = 4$$

Solution:

1. Multiply the second equation by 2:

$$2(x + 2y) = 2(4)$$

$$2x + 4y = 8$$

2. Both equations are now identical
3. These are the same line, so every point on the line is a solution

Answer: Infinitely many solutions (the equations represent the same line)

3 5 Word Problems with Full Solutions

3.1 Problem 1: Ticket Pricing

Problem Statement:

A theater sells adult and child tickets. Adult tickets cost \$6 each and child tickets cost \$3 each. Ticket sales totaled \$135. There were 25 tickets sold in total. How many adult tickets and how many child tickets were sold?

Setting up the System:

Let a = number of adult tickets and c = number of child tickets.

$$6a + 3c = 135 \quad (\text{total revenue})$$

$$a + c = 25 \quad (\text{total tickets})$$

Solution:

1. From the second equation, solve for c :

$$c = 25 - a$$

2. Substitute into the first equation:

$$6a + 3(25 - a) = 135$$

$$6a + 75 - 3a = 135$$

$$3a + 75 = 135$$

$$3a = 60$$

$$a = 20$$

3. Find c :

$$c = 25 - 20 = 5$$

4. Check: $6(20) + 3(5) = 120 + 15 = 135$ and $20 + 5 = 25$

Answer: 20 adult tickets and 5 child tickets

3.2 Problem 2: Subscription Plans

Problem Statement:

A company offers two subscription plans: Premium at \$15 per month and Basic at \$8 per month. In one month, 12 subscriptions were sold for a total of \$138. How many Premium and Basic subscriptions were sold?

Setting up the System:

Let p = number of Premium subscriptions and b = number of Basic subscriptions.

$$\begin{aligned}15p + 8b &= 138 && \text{(total revenue)} \\p + b &= 12 && \text{(total subscriptions)}\end{aligned}$$

Solution:

1. From the second equation:

$$b = 12 - p$$

2. Substitute into the first equation:

$$\begin{aligned}15p + 8(12 - p) &= 138 \\15p + 96 - 8p &= 138 \\7p + 96 &= 138 \\7p &= 42 \\p &= 6\end{aligned}$$

3. Find b :

$$b = 12 - 6 = 6$$

4. Check: $15(6) + 8(6) = 90 + 48 = 138$ and $6 + 6 = 12$

Answer: 6 Premium subscriptions and 6 Basic subscriptions

3.3 Problem 3: Distance/Rate/Time

Problem Statement:

A fast car and a slow car travel toward each other on a highway. Together, they cover 100 miles. The difference in distances they travel is 20 miles (the fast car travels 20 miles more). How far does each car travel?

Setting up the System:

Let f = distance traveled by the fast car and s = distance traveled by the slow car.

$$\begin{aligned}f + s &= 100 && \text{(total distance)} \\f - s &= 20 && \text{(difference)}\end{aligned}$$

Solution:

1. Add the two equations to eliminate s :

$$\begin{aligned}(f + s) + (f - s) &= 100 + 20 \\2f &= 120 \\f &= 60\end{aligned}$$

2. Substitute back:

$$60 + s = 100$$

$$s = 40$$

3. Check: $60 + 40 = 100$ and $60 - 40 = 20$

Answer: The fast car travels 60 miles and the slow car travels 40 miles

3.4 Problem 4: Investment

Problem Statement:

An investor deposits a total of \$5,000 in two accounts. One account earns 3% interest annually and the other earns 5% interest annually. If the combined annual interest is \$190, how much is invested in each account?

Setting up the System:

Let x = amount invested at 3% and y = amount invested at 5%.

$$x + y = 5000 \quad (\text{total investment})$$

$$0.03x + 0.05y = 190 \quad (\text{total interest})$$

Solution:

1. From the first equation:

$$x = 5000 - y$$

2. Substitute into the second equation:

$$0.03(5000 - y) + 0.05y = 190$$

$$150 - 0.03y + 0.05y = 190$$

$$150 + 0.02y = 190$$

$$0.02y = 40$$

$$y = 2000$$

3. Find x :

$$x = 5000 - 2000 = 3000$$

4. Check: $3000 + 2000 = 5000$ and $0.03(3000) + 0.05(2000) = 90 + 100 = 190$

Answer: \$3,000 invested at 3% and \$2,000 invested at 5%

3.5 Problem 5: Mixture

Problem Statement:

A chemist mixes two solutions to create a 15-liter mixture. One solution is 15% pure and the other is 30% pure. The final mixture is 20% pure. How many liters of each solution are used?

Setting up the System:

Let a = liters of the 15% solution and b = liters of the 30% solution.

$$a + b = 15 \quad (\text{total volume})$$

$$0.15a + 0.30b = 0.20(15) \quad (\text{pure substance})$$

Simplifying the second equation:

$$0.15a + 0.30b = 3$$

Solution:

1. From the first equation:

$$a = 15 - b$$

2. Substitute into the second equation:

$$0.15(15 - b) + 0.30b = 3$$

$$2.25 - 0.15b + 0.30b = 3$$

$$2.25 + 0.15b = 3$$

$$0.15b = 0.75$$

$$b = 5$$

3. Find a :

$$a = 15 - 5 = 10$$

4. Check: $10 + 5 = 15$ and $0.15(10) + 0.30(5) = 1.5 + 1.5 = 3$

Answer: 10 liters of the 15% solution and 5 liters of the 30% solution

4 Summary of Key Techniques

4.1 Substitution Method

When to use: When one variable is already isolated or can be easily isolated.

Steps:

1. Solve one equation for one variable
2. Substitute the expression into the other equation
3. Solve for the remaining variable
4. Back-substitute to find the other variable

4.2 Elimination Method

When to use: When the coefficients of a variable are opposites or can be made opposites by multiplying.

Steps:

1. Multiply one or both equations so that a variable's coefficients are opposites
2. Add or subtract the equations to eliminate that variable
3. Solve for the remaining variable
4. Back-substitute to find the other variable

4.3 Graphing Method

When to use: When a visual solution is helpful or when the equations are in slope-intercept form.

Steps:

1. Write both equations in slope-intercept form ($y = mx + b$)
2. Graph both lines on the same coordinate plane
3. Find the intersection point
4. Read the x - and y -coordinates of the intersection

4.4 Special Cases

No Solution: The lines are parallel (same slope, different y -intercepts). When solving algebraically, you get a contradiction.

Infinitely Many Solutions: The equations represent the same line. When solving algebraically, you get an identity (like $0 = 0$).

One Solution: The lines intersect at exactly one point (most common case).